02 · Executive Control Tower

> *Audience: executives, PMO, portfolio managers · Time: 30 minutes*
> *Prerequisite: module 01 · Screen:* `#/dashboard` *(Executive → Executive Control Tower)*

Learning objectives

Read the command center like an operator: interpret the telemetry strip, use the five KPI tiles, drill from any 3D chart into its source records, and judge data trust via the Integrity Core.

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1. The command header

At the top: the title, a **Live** badge (green chip showing when the bundle was generated — click refresh in a pinch), and a shortcut to the **Leadership Control Tower** (the slide-ready executive bundle, module 06).

> *Access note: this screen requires the* `executive.view` *permission. Users without it see a clear "Restricted area" panel with shortcuts to their own workspace — never a raw error.*

2. The live telemetry strip (dark HUD band)

Six always-current readouts, refreshed by realtime events as teammates work:

Readout	Shows	Watch for
Portfolio	Total projects + active/draft split	Portfolio growing faster than PM capacity
Health score	Portfolio-average RAG score /100	Trend below 60
EVM indices	Mean CPI / SPI across projects	Values < 0.95 turn amber
Forecast	Portfolio EAC + variance vs budget	Rose = forecast overrun
Alerts	Open governance alerts + overdue milestones	Any non-zero number
Governance	Pending stage gates + change requests	Queue growing week over week

The footer carries the operating model: **Plan → Execute → Monitor → Control → Govern → Deliver**.

3. The five KPI tiles

Each tile is a portal — click it and you land on the record set behind it:

Tile	Click takes you to
Portfolio Projects	Project register (`#/projects`)
Current Budget	Financials (`#/financials`)
Actual Cost	Financials (`#/financials`)
Forecast (EAC)	EVM control (`#/evm`)
Portfolio CPI / SPI	EVM control (`#/evm`)

Tiles also carry a **delta chip** (e.g. ▲ x% forecast vs budget, coloured by good/bad) and a **sparkline** of the recent trend where history exists.

4. The 3D chart wall

Four bespoke SVG 3D visualisations (no plug-ins, resolution-independent):

1. **Project Health Matrix** — an extruded 3D donut of the RAG distribution (Green / Amber / Red). *Hover a segment* to spotlight it and see its percentage; *click a slice or legend entry* to open the project register filtered to that health band. The centre shows total projects; the footer shows the average health score.
2. **Portfolio Investment Matrix** — isometric 3D columns per portfolio with three faces of data: **Budget** (blue), **Actual** (cyan), **Forecast** (amber). *Hover a cluster* for the tooltip with exact values; *click* to open Portfolios. This is the fastest way to spot a portfolio where the amber forecast column rises above the blue budget column.
3. **EVM Performance Dials** — two radial 3D gauges, **CPI** and **SPI**, from the latest period close. Green ≥ 0.95, amber ≥ 0.85, red below; the dashed tick marks the 0.95 target. The caption states plainly whether corrective control is advised.
4. **Earned Value by Project** — 3D CPI/SPI columns for the worst-performing projects. *Click a cluster* to jump straight into that project's workspace.
5. **Effort Telemetry** — a perspective 3D area chart of approved hours by week. Effort is the bridge between timesheets and cost: rising hours with flat value delivery is an early EVM warning.

5. The operational queues

Below the charts, three glass panels list what needs attention today:

- **Top Risks** — highest-scored exposures with severity dots; "Open RAID" jumps to the full register.
- **Governance Alerts** — automated threshold breaches (budget, schedule, threshold crossings) with type and timestamp; "Open queue" → Governance.
- **Capacity & Milestones** — over-allocated resources with load bars (rose above 100%), plus overdue milestones with their due dates.

Then the **Governance Queue** panel: pending stage gates and change requests in flight — the two decision lists a leadership meeting runs on.

6. The Integrity Core (dark panel, bottom right)

Trust before decisions. The panel shows:

- Recent **health changes** (project moved Amber → Red, etc.),
- A live **data freshness rail** — each project's 0–100 score with a coloured bar; the worst feed dominates the score. Red (< 40) means: the data is stale, do not decide on it — chase the timesheets/actuals first,
- A link to the full **Data Integrity** screen (`#/integrity`).

7. Realtime behaviour

The whole screen subscribes to live events (`project:health`, `actuals:changed`, `alert:created`, `governance:changed`, `evm:changed`, `project:updated`). Leave the dashboard open in a meeting — when a teammate logs hours or a health score changes, the affected tiles and charts refresh themselves.

8. Exercise (10 min)

1. Log in as `ceo@pmct.io`; open **Executive Control Tower**.
2. From the telemetry strip, note the mean CPI/SPI. Are they healthy?
3. Click the **Forecast (EAC)** tile → you land on EVM control. Find the project with the largest cost variance; note its code.
4. Back on the dashboard, click the same project's cluster in **Earned Value by Project** → its workspace opens.
5. In **Project Health Matrix**, hover the Red segment (if any) and click it.
6. Scroll to the **Integrity Core** — which project has the lowest freshness score? What is its worst feed?

Discussion: you just moved from a portfolio-level number to a single project's operational record in two clicks. How would this change a monthly review meeting?

9. Key takeaways

- Six telemetry readouts = the meeting agenda; five KPI tiles = the drill points.
- Every 3D chart is interactive: hover for values, click for context.
- The Integrity Core answers "can I trust this number?" before you decide.